Geometry : Vectors — problem 12.1.18

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The diagonals of a convex quadrilateral $ABCD$ intersect in O. Show that

$$AB^2 + BC^2 + CD^2 + DA^2 = 2\left(AO^2 + BO^2 + CO^2 + DO^2\right)$$

exactly if either $AC \perp BD$ or one of the diagonals is bisected in O.